

3B2 FPGA Programming Lab Report

Liu Zihe^{1*}

¹University of Cambridge

Abstract

This report investigates random number generation and statistical density estimation. We develop theoretical foundations for histogram uncertainty using multinomial distribution theory, implement the inverse CDF transformation method for generating non-uniform random variables, and characterize heavy-tailed α -stable distributions. Empirical validation demonstrates convergence rates for Monte Carlo estimation and the advantage of kernel density estimation over histograms.

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References

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